12–15 Minute Project Presentation Structure

1. Title Slide (30 seconds)

Include:

- Project title
- Course: **CST8219 – C++ Programming**
- Student name and ID
- Instructor name
- Date of presentation

2. Problem Statement / Motivation (1 minute)

Explain:

- What problem the system solves
- Why the system is useful

Keep this very brief.

3. System Overview (1–2 minutes)

Explain the main features of the system. A simple system diagram or architecture diagram helps.

4. Software Design (2–3 minutes)

Explain the core class structure.

Show:

- UML Class Diagram

Highlight:

- Main classes
- Relationships
- Inheritance

Explain **why the design was chosen**.

5. Key C++ Concepts Used (2–3 minutes)

Students must highlight course concepts, such as:

Object-Oriented Programming

- Classes
- Inheritance
- Polymorphism
- Abstract classes

C++ Features

- Templates
- Operator overloading
- Friend functions
- Smart pointers

STL Containers

- vector
- map
- unordered_map

Explain how they were used in the project.

6. GUI Demonstration (Qt) (3–4 minutes)

Show a live demo or screenshots.

Demonstrate:

- Main window
- Adding records
- Performing operations
- Viewing analytics/results

Focus on:

- User interaction

- Input validation
- Output display

7. Data Storage (1 minute)

Explain briefly:

- How data is saved
- File format used (JSON / text)
- Loading data on startup

8. Testing and Edge Cases (1 minute)

Show examples of:

- Invalid input
- Error handling
- Boundary conditions

9. Challenges and Solutions (1 minute)

Explain:

- Major difficulties
- How they were solved

Examples:

- Qt GUI integration
- Memory management
- File handling

10. Conclusion (30 seconds)

Summarize:

- What the project achieved
- Key concepts learned
- Possible future improvements

Recommended Slide Count

Section	**Slides**
Title	1
Problem	1
System Overview	1
Design / UML	1–2
C++ Concepts	1–2
GUI Talks	2
Testing	1
Challenges	1
Conclusion	1